SHOCK & ISOLATION / ABOUT 10 MINUTES

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When a soft mount is not a soft system

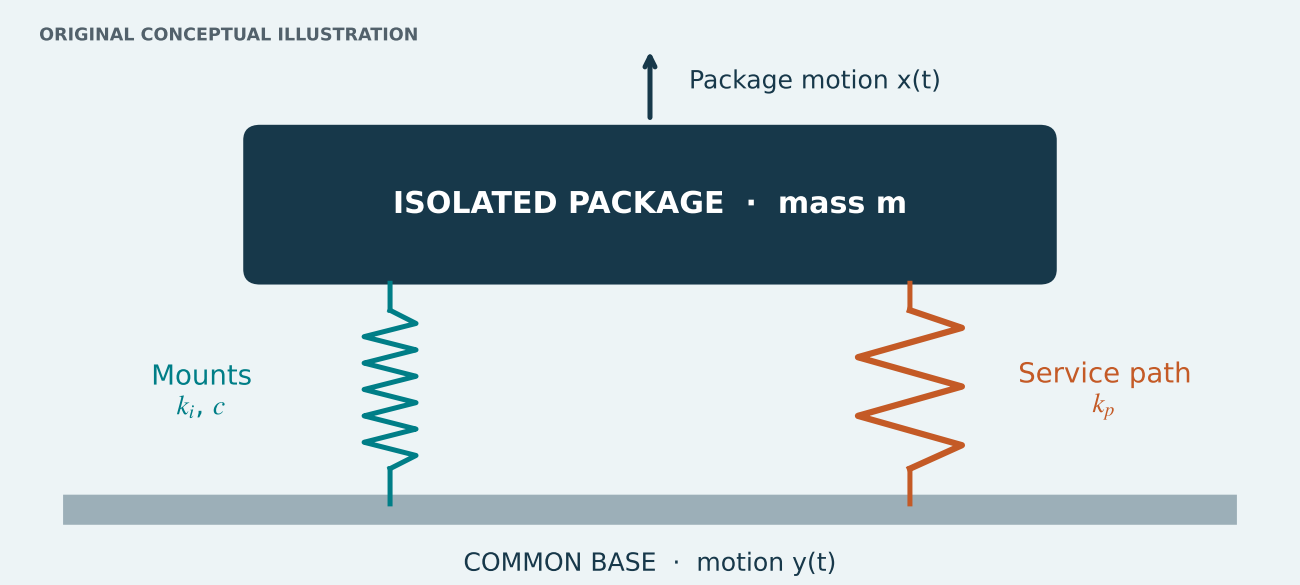


Figure 1. Two mechanical routes connect the same base to the package. The service path represents an equivalent stiffness of a tube, harness, or strap—not a documented flight installation. Original illustration.

A package sits on compliant mounts, yet its installed resonance is much higher than the mount calculation predicts. The mounts may be exactly as specified. The missing spring may be the tube entering the housing, the tightly dressed harness, or a strap whose purpose was electrical or thermal.

Today’s objective is to recognize those parallel transmission paths, estimate their effect on shock response, and decide what evidence is needed before crediting an isolation system. The key question is not whether a mount is soft. It is whether the complete installed connection lets the package move in the directions that matter.

Start with a deliberately small model. It makes one counterintuitive result visible: an assembly can move less relative to its base while experiencing more acceleration. Reduced motion alone is not proof of improved protection.

01 / FOLLOW EVERY MECHANICAL CONNECTION

The spring you did not buy

A service connection is also a structural connection. Its stiffness depends on geometry, attachment constraints, and direction; “flexible cable” is not a usable boundary condition. NASA’s g-LIMIT work measured candidate umbilical forces and moments, examined looping and mounting arrangements, and derived stiffness matrices for isolation analysis. Those were microgravity experiments, not pyroshock qualification tests, but the modeling lesson transfers directly [2].

For a first screen, let a rigid package of mass m translate along one axis. Every connection sees the same rigid-base motion y(t). Its absolute motion is x(t); relative travel is z = x − y. Add the mount stiffness k_i and parallel-path stiffness k_p. Let c be the total viscous damping coefficient:

$$m\ddot{z} + c\dot{z} + (k_i + k_p)z = -ma_b(t)$$

Here base acceleration a_b has units m/s², stiffness is N/m, damping is N·s/m, and travel is metres. This is the standard base-excited oscillator [1], with an additional parallel spring. The installed frequency and damping ratio become:

$$f_n = \frac{1}{2\pi}\sqrt{\frac{k_i + k_p}{m}} \qquad \zeta = \frac{c}{2\sqrt{m(k_i + k_p)}}$$

If the added path contributes stiffness but negligible damping, it raises frequency and lowers the damping ratio. Keeping the original damping ratio in a model would silently add damping that the hardware does not possess. A real service path may dissipate energy too; then characterize and include that contribution.

The scalar sum is only a screening model. Offset attachments produce moments, so a six-degree-of-freedom rigid-body model may be the next useful step. Flexible decks, tubing modes, and local equipment modes demand still more coordinates. Connections to different moving supports require distinct inputs; their effects cannot be captured by one common base acceleration.

02 / A CONTROLLED NUMERICAL COMPARISON

Twice the frequency, nearly twice the peak

Consider a generic 2.0 kg instrument package. Its mounts alone have a 100 Hz natural frequency and damping ratio 0.10. We add a purely elastic bypass with three times the mount stiffness. These are chosen demonstration values, not a reconstruction of any vehicle, valve, or flight environment.

Model quantity	Mounts only	With bypass
Total stiffness	0.790 MN/m	3.158 MN/m
Damping coefficient	251 N·s/m	251 N·s/m
Natural frequency	100 Hz	200 Hz
Damping ratio	0.10	0.05
Peak absolute acceleration	67.8 g	127.9 g
Peak relative travel	1.651 mm	0.790 mm

Table 1. Computed responses to the same 100 g, 2 ms half-sine base-acceleration pulse; g = 9.80665 m/s². Initial relative displacement and velocity are zero. No measured data.

Both models receive one positive half-sine pulse, followed by zero acceleration. We integrate the linear state equations for 100 ms using a 5 microsecond step. Halving the step changes both reported response peaks by less than 0.1%. The half-sine is an idealized mechanical-shock input; its nonzero velocity increment and short duration are not claims about a separation event.

The acceleration reported is the package’s absolute acceleration, calculated from the total connection force:

$$a_p = -\frac{c\dot{z} + (k_i + k_p)z}{m}$$

The bypass carries a peak elastic force of about 1.87 kN in this model. That force is not a connector allowable, and it does not include local bending or dynamic amplification within the service connection. It identifies a load path that deserves a real structural model.

03 / TWO METRICS, TWO DIFFERENT STORIES

Isolation buys motion

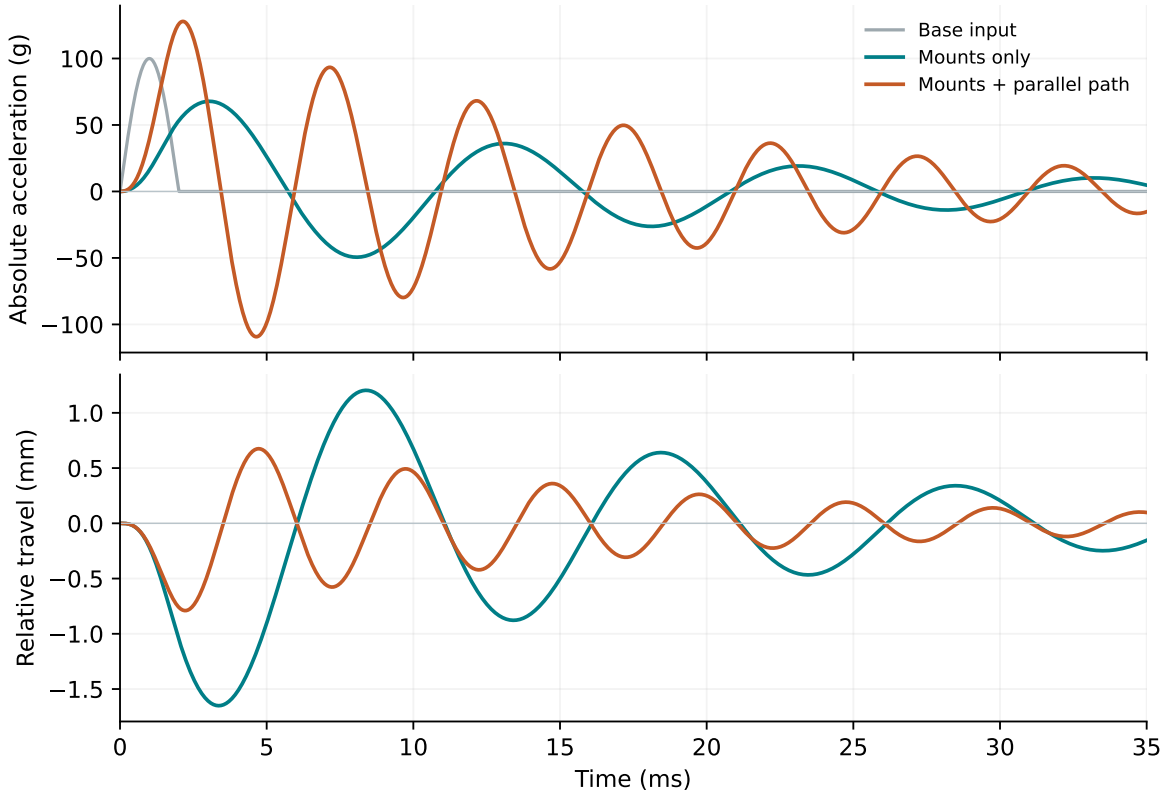


Figure 2. Original simulation of the Table 1 models; first 35 ms shown. Upper panel: absolute package acceleration. Lower panel: package displacement relative to the base. The input ends at 2 ms; ring-down continues.

With only the mounts, the package lags farther behind the base and experiences a smaller acceleration peak. The added connection restrains that lag, but transmits a larger force. For this pulse, the price of lower travel is an acceleration peak almost 1.9 times larger. Another pulse duration or waveform can change the ratio; this is not a universal penalty factor.

Travel still matters. If the softer arrangement reaches a stop, touches a nearby panel, or pulls a harness taut, a new load path appears during the event. The linear prediction then ceases to describe the hardware. Clearance must cover tolerances and the relevant combined motions, not just the nominal single-axis displacement plotted here.

A familiar harmonic rule also needs care. In the linear viscous single-mode model, acceleration transmissibility drops below unity only above √2 times the installed natural frequency [1]. That crossover is not a shock acceptance rule. A transient excites a band of frequencies, and multiplying an input SRS by a transmissibility magnitude does not generally produce the output SRS. Propagate a defensible time history through the dynamic model, then calculate the output SRS.

04 / MAKE THE ANALYSIS MATCH THE ASSEMBLY

Draw the boundary before choosing the mount

First identify the failure-sensitive response: housing acceleration, internal-board strain, connector load, relative travel, or moment at a port. A lower housing SRS does not automatically establish acceptable internal response. State the configuration at the event, including fluid inventory or pressure where relevant, cable dressing, brackets, restraints, and available clearance. These are model inputs.

Then inventory every crossing of the isolation boundary. For each, determine attachment locations, orientation, preload, and how much motion is available before contact or straightening. Where uncertainty dominates, measure force versus displacement in the relevant directions. Use dynamic characterization when frequency-dependent behavior matters. A static stiffness is a useful start, not a guarantee that the connection stays spring-like throughout the shock band.

In an FEM, represent the package mass and inertia, mount locations, service paths, and supporting structure. Compare a mounts-only model with the fully connected model. Inspect changed mode shapes and interface forces as well as frequencies. Sweep uncertain path stiffness, damping, attachment constraints, and clearances. A reduced model is credible only over the frequency range where omitted flexibility does not control the result.

An OSIRIS-REx lesson shows why transmission paths deserve this attention. The project reported that updated analyses and testing found a pressure wave from drogue-parachute mortar firing interacting directly with the sample-return capsule avionics deck. The shock was reported as ten times higher than expected. Wire-rope isolation was developed within tight volume constraints, with harness changes needed for the increased standoff [3]. This was a documented pressure-wave coupling issue—not evidence that a harness bypass caused the shock.

Finally, test the installed configuration with representative connections and restraints. Measure both sides of the isolation boundary and check for contact, level dependence, and functional interruption. Compare time histories and relevant spectra with the analysis; investigate discrepancies before adjusting model parameters. NASA’s 2024 discussion emphasizes assembly testing to validate unit environments and cautions that propagation depends on joints and configuration [4].

NASA-STD-7003B, as discussed in that article, concerns pyroshock test criteria for NASA hardware. The article also proposes revisions; proposals are not adopted requirements. Use the program’s governing revision and tailoring. Nothing in this lesson establishes a universal attenuation target, allowable stroke, or qualification margin.

05 / TAKE IT INTO THE DESIGN REVIEW

Three things to carry forward

1. Count paths, not mounts. Tubes, harnesses, and straps can determine installed stiffness and introduce moments. Their intended function does not remove their mechanical role.

2. Track acceleration and travel together. In the worked example, the bypass cuts travel by roughly half while raising peak acceleration from 68 g to 128 g. Protection and clearance are coupled decisions.

3. Validate the connected assembly. A catalog mount curve or a disconnected test cannot establish the response of the final installation. Include uncertain paths and possible contact in the evidence plan.

A short reflection

A supplier demonstrates 100 Hz on a representative mass. Your installed assembly measures 200 Hz. Assuming the same mass, unchanged mount stiffness, one translational mode, and negligible damping effects on the identified frequency, what equivalent extra stiffness does that suggest?

Answer: Total stiffness has increased by a factor of four, so the added path contributes approximately three times the mount stiffness. Treat that as a diagnostic estimate. Different mode shapes, support flexibility, or rotational coupling could invalidate the inference; inspect the mode before blaming a specific connection.

Sources and figure notes

[1] M. Whorton, Fundamentals of Microgravity Vibration Isolation, NASA MEIT-2003, Section 14, March 6, 2003, pp. 8–10. Base-excited oscillator and damping trade.

[2] R. C. Engberg, Umbilical Stiffness Matrix Characterization and Testing for Microgravity Science Payloads, AIAA 2003-1873, 2003. NTRS abstract: loops, displacement range, attachment geometry, measured stiffness matrices.

[3] D. Everett et al., Designing to Sample the Unknown: Lessons from OSIRIS-REx Project Systems Engineering, 2017, printed p. 12. SRC pressure-wave coupling and wire-rope isolation redesign.

[4] NASA, Mentoring the Next Generation of Engineers and Improving Shock Testing Standards, March 21, 2024, “Updating Guidance…” section. Distinguishes existing NASA-STD-7003B provisions from proposed revisions.

Figures 1–2 and Table 1 are original illustrations/calculations for this lesson. The opening image is a conceptual schematic, not a historical photograph. All numerical properties are illustrative. Source links checked October 17, 2026.

Model note: this run could not verify or select GPT-6 Astra.